

Parth Sarthi

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Summary

Computer Science graduate with hands-on experience in Artificial Intelligence, Machine Learning, NLP, and Full-Stack AI application development. Skilled in building production-oriented ML systems, REST APIs, data pipelines, and interactive web applications using Python, Flask, TensorFlow, and cloud-oriented architectures. Experienced in autonomous vehicle simulation research, AI dashboards, and scalable GitHub-based engineering workflows.

Education

B.Tech in Computer Science & Engineering (2021-2025)

Sharda University

Technical Skills

- **AI / ML:** Python, NumPy, Pandas, TensorFlow, PyTorch, Keras, Matplotlib, NLTK, NLP, Scikit-learn
- **Computational Chemistry:** GROMACS, XGrace, CHARMM36, Avogadro
- **Web Development:** React, HTML, CSS, JavaScript, Flask, REST APIs
- **Languages:** Python, Java, SQL, Bash
- **Tools & Platforms:** GitHub, VS Code, Jupyter Notebook, Tableau, Postman, Git, AWS (S3, IAM), IntelliJ

Experience

Web Developer Intern

Bharat Intern

May 2023 – June 2023

- Developed and deployed three independent web applications using HTML, CSS, and JavaScript.
- Published projects as separate GitHub repositories with live deployments.
- Projects included a portfolio website, Netflix homepage clone, and a temperature converter web app.

Projects

Self-Driving Car Simulation Web Dashboard

Python, CARLA Simulator, WebGL, Three.js, HTML, CSS, JavaScript

- Built an interactive AI-driven dashboard to visualize and learn autonomous vehicle simulations.
- Implemented a novel terrain self-adaptation method for autonomous vehicles.
- Research outcome led to a government-certified patent.

Pharmacokinetics, Molecular Docking & Dynamic Simulation

Python, Bash, GROMACS, XGrace, Matplotlib

- Worked on protein-ligand complex simulations in collaboration with Delhi University researchers.
- Handled scripting, simulation execution, and post-simulation analysis.

AI Resume Analyzer & ATS Scoring System

Python, Flask, NLP, PDFPlumber, HTML, CSS, JavaScript, REST API, GitHub, AWS S3

- Built a full-stack AI-powered Resume Analyzer and ATS Scoring System capable of extracting, analyzing, and matching resume skills against job descriptions.
- Developed an NLP-based skill extraction and ATS scoring pipeline using Python and PDF text extraction techniques to identify matched and missing skills.
- Implemented RESTful Flask APIs for resume upload, analysis, and JSON-based communication between frontend and backend systems.
- Designed an interactive frontend dashboard for resume upload and real-time ATS score visualization.
- Structured the project using production-style modular backend architecture including API, extraction, NLP, and scoring layers with GitHub-based version control workflows.
- Integrated cloud-oriented architecture planning using AWS services including S3 and IAM for scalable deployment readiness.

Spam Email Detection Using Machine Learning

Python, TensorFlow, Jupyter Notebook, NumPy, Pandas, Matplotlib, Seaborn, NLTK, WordCloud

- Designed and trained a supervised machine learning model in Jupyter Notebook to classify emails as spam or legitimate using a labeled mixed-email dataset.
- Implemented text preprocessing and feature extraction using NLP techniques (tokenization, stop-word removal, vectorization).
- Achieved **97.3% classification accuracy** in spam detection through model optimization and evaluation.

Achievements & Awards

- Research Paper: “Self-Driving Car Simulation Using Machine Learning” (Pages 219-229) Published at **Data Mining and Information Security Proceedings of ICDMIS 2025, Volume 3 Book Series**.
- Indian Government Certified Patent: *Terrain Adaptive System for Autonomous Vehicles*
Patent No: 202511017309